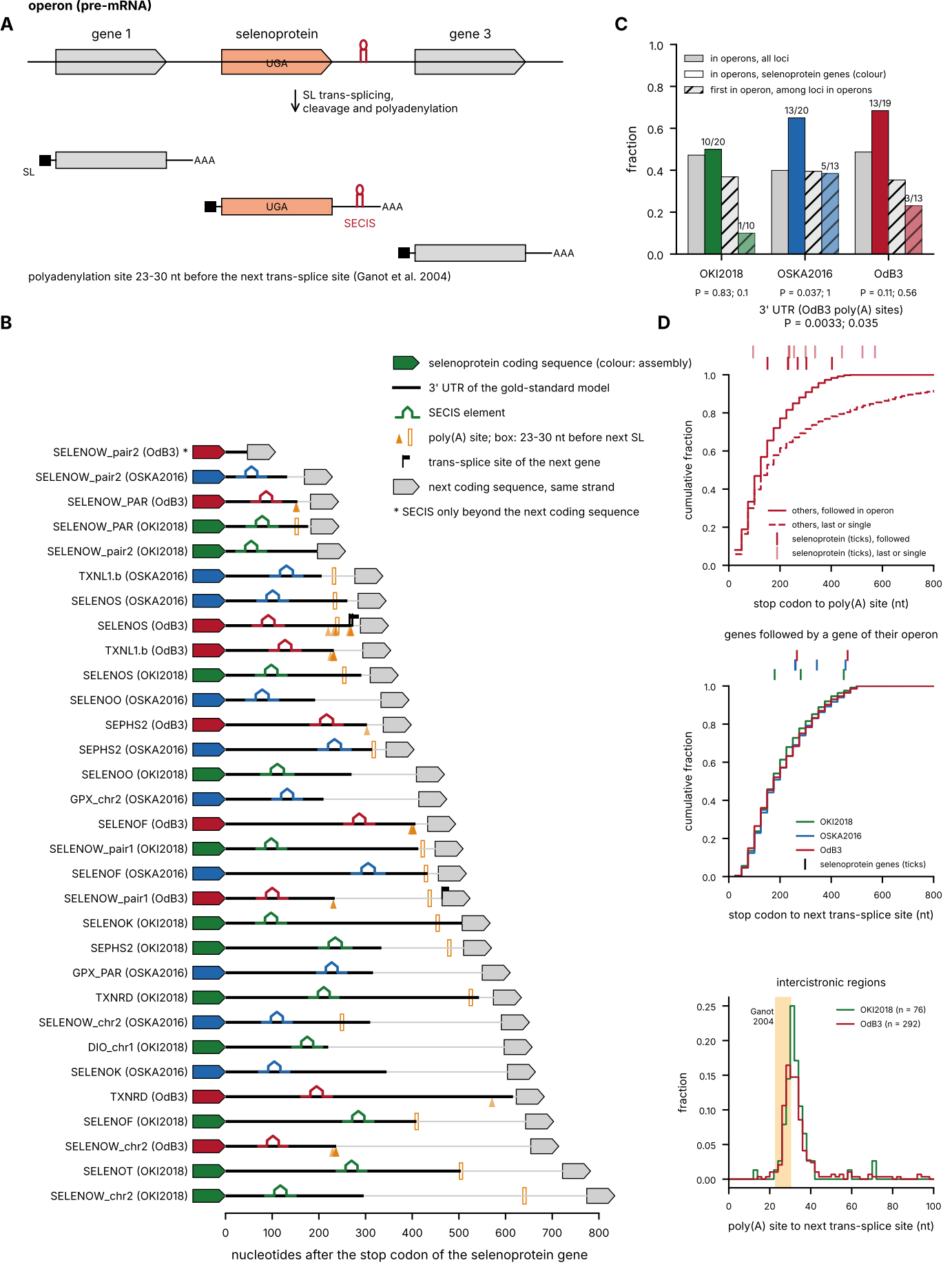
A
operon (pre-mRNA)
gene 1
selenoprotein
gene 3
UGA
SL trans-splicing,
cleavage and polyadenylation
SL
AAA
UGA
AAA
SECIS
AAA
polyadenylation site 23-30 nt before the next trans-splice site (Ganot et al. 2004)
C
in operons, all loci
in operons, selenoprotein genes (colour)
first in operon, among loci in operons
fraction
10/20
1/10
13/20
5/13
13/19
3/13
OKI2018
OSKA2016
OdB3
P = 0.83; 0.1
P = 0.037; 1
P = 0.11; 0.56
3' UTR (OdB3 poly(A) sites)
P = 0.0033; 0.035
B
selenoprotein coding sequence (colour: assembly)
3' UTR of the gold-standard model
SECIS element
poly(A) site; box: 23-30 nt before next SL
trans-splice site of the next gene
next coding sequence, same strand
* SECIS only beyond the next coding sequence
SELENOW_pair2 (OdB3) *
SELENOW_pair2 (OSKA2016)
SELENOW_PAR (OdB3)
SELENOW_PAR (OKI2018)
SELENOW_pair2 (OKI2018)
TXNL1.b (OSKA2016)
SELENOS (OSKA2016)
SELENOS (OdB3)
TXNL1.b (OdB3)
SELENOS (OKI2018)
SELENOO (OSKA2016)
SEPHS2 (OdB3)
SEPHS2 (OSKA2016)
SELENOO (OKI2018)
GPX_chr2 (OSKA2016)
SELENOF (OdB3)
SELENOW_pair1 (OKI2018)
SELENOF (OSKA2016)
SELENOW_pair1 (OdB3)
SELENOK (OKI2018)
SEPHS2 (OKI2018)
GPX_PAR (OSKA2016)
TXNRD (OKI2018)
SELENOW_chr2 (OSKA2016)
DIO_chr1 (OKI2018)
SELENOK (OSKA2016)
TXNRD (OdB3)
SELENOF (OKI2018)
SELENOW_chr2 (OdB3)
SELENOT (OKI2018)
SELENOW_chr2 (OKI2018)
0
100
200
300
400
500
600
700
800
nucleotides after the stop codon of the selenoprotein gene
D
cumulative fraction
others, followed in operon
others, last or single
selenoprotein (ticks), followed
selenoprotein (ticks), last or single
stop codon to poly(A) site (nt)
genes followed by a gene of their operon
cumulative fraction
OKI2018
OSKA2016
OdB3
selenoprotein genes (ticks)
stop codon to next trans-splice site (nt)
intercistronic regions
Ganot
2004
OKI2018 (n = 76)
OdB3 (n = 292)
fraction
poly(A) site to next trans-splice site (nt)